

LIET – 30 Days In-campus Training
Subject – Data Analytics

Assignment 4: 23rd February, 2026

Topic – NumPy, Pandas, SQL

Assignment 4:

1. Write an SQL query to display the names and salaries of employees whose salary is greater than the average salary in the company.
Explain how the subquery works.
2. Write a query to retrieve the top 5 highest-paid employees from an employees table.
Explain how sorting affects the output.
3. Write a query to calculate:
 - Total number of employees
 - Average salary
 - Minimum and maximum salaryExplain the difference between aggregate and scalar functions.
4. Given a sales table with columns (region, amount), write a query to find total sales per region.
Filter only those regions where total sales exceed 50,000.
5. Write a query to find the number of unique job roles in an employees table.
Explain why DISTINCT is necessary here.
6. Write a query to retrieve students who scored between 60 and 80 marks.
Rewrite the same query using BETWEEN.

7. Write a query to display employees whose commission is NULL.
Explain the correct way to check NULL values in SQL.
8. Write a query to increase the salary of employees in the "IT" department by 10%.
Explain how arithmetic operations are handled in SQL.
9. Write a query to delete records of students who scored less than 40 marks.
What precaution should be taken before executing DELETE?
10. Write a query to find employees who earn more than the average salary of their department (without using joins).
Explain the logic of the subquery.

Note: This assignment is meant strictly for your own learning and practice. Students are advised **not to copy answers from any website or online platform**. Attempt all questions independently to gain maximum understanding.

Assignment submission form link - <https://forms.gle/ccNkKz2BuaVyqiDE8>

